PLASMA PHYSICS & TOKAMAK CONFINEMENT — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 15, 2026

Diagram-based questions on Plasma Physics & Tokamak Confinement

Question 1

10.0 points

Examine the toroidal and poloidal magnetic field geometry in the tokamak cross-section depicted in the diagram below. Discuss how these fields collaborate to confine plasma, and identify any potential areas of instability within the configuration.

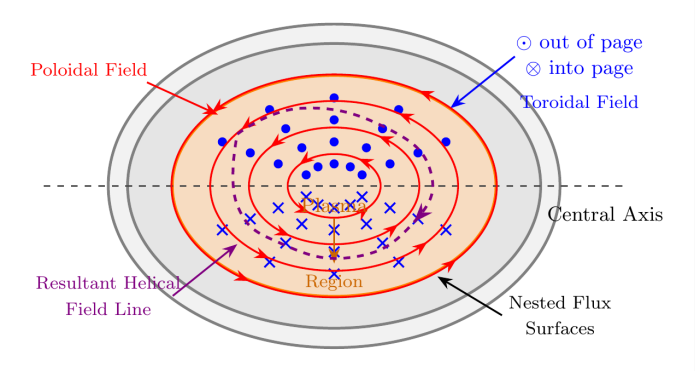

Fig. 1. A cross-sectional diagram of a tokamak showing the toroidal and poloidal magnetic fields. The toroidal field lines should be depicted as concentric circles around the central axis of the tokamak, while the poloidal field lines should be shown as loops encircling the plasma vertically. The plasma region should be highlighted in the center of the tokamak, and potential areas of instability, such as regions where the field lines are closely spaced or intersect, should be indicated. Labels should include 'Toroidal Field', 'Poloidal Field', and 'Plasma'.

Question 2

10.0 points

Interpret the Lawson criterion diagram below that displays the triple product $nT\tau$ versus temperature. Explain how this diagram is used to evaluate the feasibility of fusion reactions.

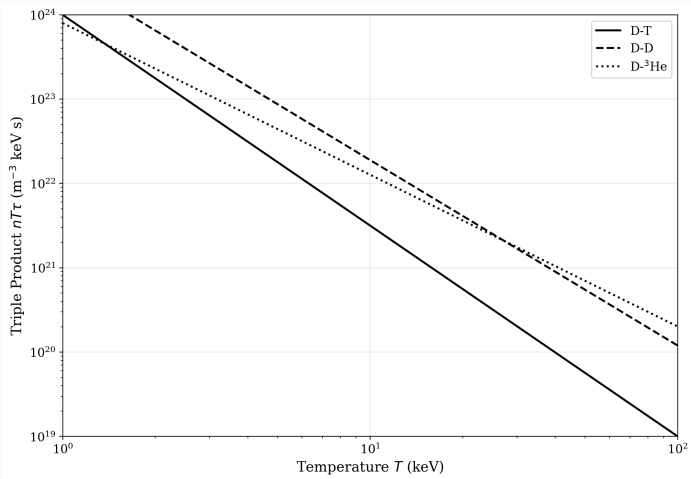

Fig. 2. Lawson criterion plot showing the triple product nTτ versus temperature for fusion reactions. The plot includes regions indicating feasibility for different fusion reactions, such as D-T and D-D, with labeled axes for temperature and nTτ. The diagram highlights areas where the triple product meets the conditions necessary for sustained fusion. IMPORTANT correction needed: The diagram description explicitly states the diagram should show. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 3

10.0 points

Examine the diagram below illustrating magnetic field curvature and grad-B drift for a charged particle in a tokamak. Determine how these drifts impact particle confinement and suggest mitigation strategies.

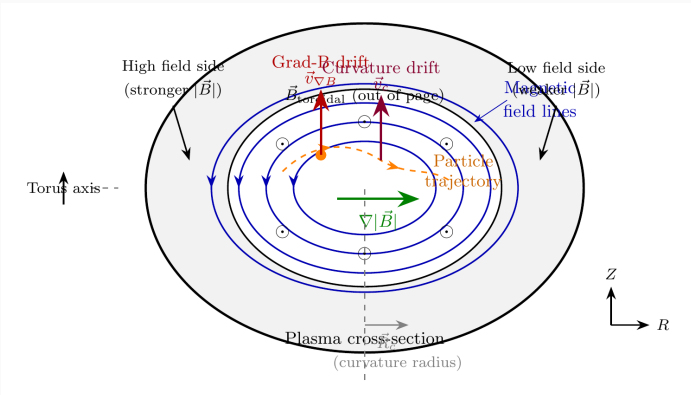

Fig. 3. A diagram showing magnetic field lines in a tokamak with a charged particle experiencing curvature drift and grad-B drift. The tokamak is represented as a toroidal shape with magnetic field lines curving around it. A charged particle is shown moving along these field lines, with arrows indicating the direction of curvature drift and grad-B drift. The magnetic field lines should be labeled, and the drifts should be indicated with arrows and labels to show their directions. The tokamak should be depicted in a cross-sectional view to highlight the curvature of the magnetic field lines and the resulting particle drifts.

Question 4

10.0 points

Examine the q-profile (safety factor) and magnetic flux surfaces in the tokamak depicted in the diagram below. Discuss how the shape of the q-profile influences plasma stability and confinement.

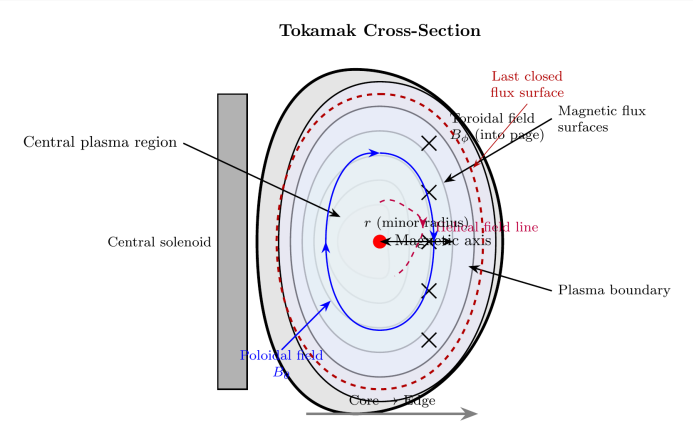

Fig. 4. A cross-sectional diagram of a tokamak showing magnetic flux surfaces and the q-profile. The diagram should include concentric magnetic flux surfaces labeled with their respective q-values (safety factor). The central plasma region should be highlighted, and the magnetic field lines should be depicted as loops around the central axis. The q-profile should be represented as a curve within the tokamak, illustrating how the safety factor changes from the core to the edge of the plasma. Include labels for key components such as the central solenoid, plasma boundary, and magnetic field lines. IMPORTANT: The diagram is missing required labels. The diagram is missing the q-profile curve and the. Add each missing label as visible text at the appropriate location in the diagram. IMPORTANT: The diagram is missing required labels. The diagram is missing the q-profile curve and. Add each missing label as visible text at the appropriate location in the diagram.

Question 5 10.0 points

Review the diagram below illustrating plasma instability growth rate as a function of wave number. Explain the implications of this relationship for tokamak operation and stability control.

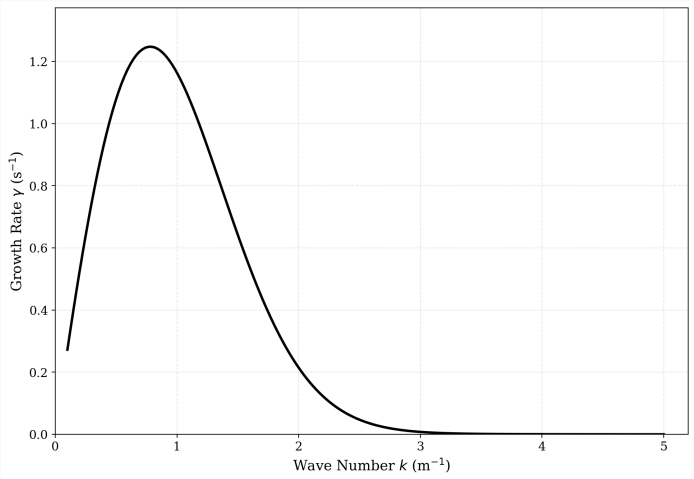

Fig. 5. A plot showing plasma instability growth rate as a function of wave number. The x-axis represents the wave number, and the y-axis represents the growth rate. The curve should illustrate how the growth rate changes with varying wave numbers, highlighting any key features such as peaks or thresholds that are relevant for tokamak operation and stability control.